



Armada Kalamunda Group

Postpartum Haemorrhage Guideline

1. Purpose

- To identify women at increased risk of Postpartum Haemorrhage (PPH) and plan appropriate prophylactic measures and early interventions if required.
- To identify and manage the cause of PPH and control excessive bleeding, minimising the risk of morbidity and mortality.

2. Applicability

This guideline applies to the medical and midwifery discipline within the maternity department of Armada Health Service.

3. Guidelines

- PPH is potentially a life-threatening complication of vaginal and caesarean delivery. It can result in significant maternal morbidity and remains one of the leading causes of maternal mortality in Australia and internationally.
- The incidence of PPH is thought to be underestimated by as much as 50%, due to the clinical difficulty in accurately estimating blood loss (EBL).
- Approximately two thirds of the incidence of PPH cannot be predicted. The risk of submitting a woman to unnecessary interventions and subsequent iatrogenic disease must also be weighed against the individual's risk of PPH.

- **Definition of PPH:**

Traditionally PPH has been defined as a blood loss of 500ml or more during puerperium. Further classification of PPH into primary (within 24 hours of delivery) and secondary (between 24 hours and six weeks postpartum) is also well established.

3.1 Classifications of PPH

- Mild: 500 -1000mls (in the absence of clinical signs of shock)
- Moderate: 1000-2000mls
- Severe: 2000mls and above

3.2 Aetiology of Primary PPH 'The 4 T's' to aid with diagnosis.

3.2.1 Tone – abnormalities of uterine contraction (the most common cause-70%)

- Atonic Uterus
- Over distended uterus
- Uterine muscle exhaustion
- Intra-amniotic/uterine infection

- Drug induced hypotonia
- Functional or anatomical distortion of the uterus

3.2.2 Trauma – genital tract trauma-20%

- Episiotomy or lacerations (Cervix, vagina or perineum)
- Extensions, lacerations at Caesarean section
- Uterine rupture
- Uterine inversion

3.2.3 Tissue – retained products of conception

- Retained Placenta
- Abnormal Placenta
- Retained cotyledon or succenturiate lobe

3.2.4 Thrombin – abnormalities of coagulation-1%

- Coagulopathies
- Disseminated intravascular Coagulation
- Pre-eclampsia/HELLP syndrome
- Therapeutic anti coagulation

3.3 Clinical Risk Factors for Primary PPH

3.3.1 Antepartum

- Anaemia
- Extremes of age
- Primigravida
- BMI >35
- Existing, or pregnancy acquired coagulation disorders
- Pre-eclampsia
- Antepartum haemorrhage including abruption
- Past history of PPH
- Past history of retained placenta and /or membranes
- Past history of manual removal of placenta and/or membranes
- Previous uterine surgery
- Over distended uterus (multiple birth, polyhydramnios, uterine fibroids)
- Grand multiparity
- Asian descent or Hispanic ethnicity
- Low lying placenta

3.3.2 Intrapartum

- Prolonged labour
- Prolonged first stage
- Physiological management of 3rd stage
- Precipitous or in co-ordinate labour and / or birth

- Operative births vacuum, forceps and caesarean section)
- Induction and/ or augmentation
- Drug therapy (magnesium sulphate, terbutaline, Nifedipine, anti-coagulants
- Placental abruption
- Abnormal placental insertion
- Anatomic distortions of the uterus
- Large baby / large placental site
- Delay in delivery of placenta
- Episiotomy or cervical tear
- Chorioamnionitis
- Taking selective serotonin reuptake inhibitor (SSRI) or serotonin-noradrenaline reuptake inhibitor (SNRI) antidepressants in the month before birth may result in a small increased risk of postpartum haemorrhage,

3.4 Management of women identified as being at clinical Risk of PPH

Prophylactic therapy and classification of women according to clinical risk factors is not a substitute for prevention and close observation of every woman for PPH post birth.

Preventative management in those patients identified as high risk includes:

- Antenatal management of iron deficiency anaemia
- In Labour:
- Insert an intravenous cannula (IVC) (16G or larger)
- Collect and send blood for group and hold, full blood picture and consider cross matching blood
- For active management after vaginal birth:
 - Administer 10 units of oxytocin (by intramuscular injection) or
 - 5 units of oxytocin slow intravenous injection over 3 to 5 minutes for women who have received oxytocin during labour or
 - 5 units of oxytocin plus 500 micrograms of ergometrine (by intramuscular injection) immediately after the birth of the baby and before the cord is clamped and cut
- Commence prophylactic infusion of Oxytocin (40 units of Oxytocin in 500ml compound sodium lactate solution) at 125ml/hour preferably commenced after delivery of the placenta and membranes. Increase to 250ml/hour if bleeding continues and decrease by 40 ml/hour every 30 minutes to titrate down when bleeding is controlled.
- In addition to uterotonic drugs, give tranexamic acid (1 g by intravenous injection over 10 minutes). Repeat if necessary after at least 30 minutes for managing continuing postpartum haemorrhage.

3.5 Management of Primary PPH (Appendix 1)

3.5.1 Initiate Emergency Care-Performed concurrently

Assign roles to staff members to enable simultaneous management, including a scribe responsible for documentation on the AKMR73.

- Press the YELLOW BELL Staff Assist button located in the room to call for assistance from the co-ordinator and other midwives on duty.

- If the woman is hemodynamically unstable, or if unable to get immediate medical help, dial 55 and call a code blue Obstetrics. Ensure the duty anaesthetist is aware of ongoing PPH > 500 ml and attends to assist with fluid management and maintenance of the airway in the advent of the deteriorating patient.

3.5.2 Resuscitation performed concurrently with identifying and treating the cause.

- Insert two intravenous cannula (IVC) preferably 16 gauge if there is any difficulty siting an IVC. Call the anaesthetist for help.
- Collect blood for Group and Hold ,FBC, Coagulation profile, ROTEM (Rotational Thrombo Elastometry).
- Consider cross matching blood (at least 2 units)
- Insert an indwelling urinary catheter (and attach a urinary collection bag with an hourly chamber).
- Monitor the maternal observations every 5 mins-SpO2, respiratory rate, blood pressure, conscious state, blood loss and uterine fundus. Administer Oxygen if necessary.
- Keep the woman warm.

3.5.3 Volume Replacement:

Initiate volume replacement with crystalloids e.g. compound sodium lactate (Hartmann's).If there is ongoing bleeding, consider O-Negative blood or group specific or cross-matched packed cells.

3.6 Massive Obstetric Haemorrhage Protocol activation:

Armada Health Service Obstetric Massive Transfusion Protocol (Appendix 2)

For any of the following:

- Haemorrhagic shock
- Estimated blood loss > 1500 ml
- Laboratory or clinical evidence of coagulopathy
- 4 units of red cells transfused and more anticipated.
- Woman is still bleeding after initial management.

Notifying the laboratory on Ext 2028/2030 as soon as possible

Identify and Treat the Cause

3.6.1 Uterine Atony

- Ongoing massage of the uterus and /or bimanual compression.
- Give a second dose of Uterotonics, e.g. Syntometrine 1 ml IM or Ergometrine 250 micrograms IM , with consideration for second 250 microgram dose IM as needed.
- Beware of women with cardiac disease or hypertensive women who may develop extreme hypertension following the administration of Ergometrine. Ergometrine should only be used after consultation with the consultant Obstetrician in the setting of pre-existing hypertension.
- Commence an Oxytocin infusion of 40 units in 500 ml compound sodium lactate at 125ml/hour. Increase to 250 ml/hour if ongoing bleeding and decrease by 40 ml/hour every 30 min to titrate down when PPH controlled.
- Carbetocin 100microgram (1ml) IM or IV slowly over 1 minute. Carbetocin is a synthetic long acting Oxytocin which can be used either therapeutically or prophylactically for PPH.

However, Carbetocin is not superior to an oxytocin infusion. Recurrent doses of Carbetocin are not indicated in ongoing PPH. Carbetocin is mostly used in Caesarean settings at AHS.

- Use of Carbetocin or Syntocinon in individual situations should be the discretion of the obstetrician in consultation with the anaesthetist. Note that after intramuscular administration, oxytocin acts in about 2.5 minutes and the effects last about 30 minutes to 1 hour. Ergometrine acts in about 7 minutes and the effects last about 3 hours
- Consider oxytocin infusion as above if bleeding continues post Carbetocin injection.
- Consider administering Tranexamic Acid 1 g IV slowly as a bolus dose if the blood loss is more than 500ml after giving other oxytocics or if there are multiple risk factors. Consider the risks and benefits of Tranexamic Acid in patients with risk factors for thromboembolic complications (ischaemic heart disease). Caution should be used in patients with known thrombophilia's.

3.6.2 If The Uterus remains atonic

- Administer Carboprost 250microg intramuscular (IM) (1 ampoule=1ml). Do not give Carboprost intra-myometrial and ensure that it is not injected intravascularly when given IM.
- **Do not use in women with cardiac or pulmonary disease, hepatic or renal insufficiency, or allergic to prostaglandins. Use with caution in women with asthma.**
- The first dose of Carboprost may be given on the labour ward in the presence of specialist Obstetrician and/ or anaesthetist and only when the estimated blood loss is reaching or exceeding >500ml and other Uterotonics have been unsuccessful.
- Approximately 72% of cases of uterine atony can be managed with a single dose of Carboprost. If the uterus remains atonic after 15 minutes, further doses of Carboprost may be given at 15 minute intervals to a maximum of 8 doses(2mg) if required, and can be given only in the operating theatres at AHS.

Use of Carboprost:

- Whenever Carboprost has been administered as part of the management of a PPH, a Category A Form Special Access Scheme (SAS) must be completed and sent to AHS pharmacy.
- Check the completeness of the placenta and membranes.
- If bleeding is more than 1000ml and ongoing, transfer the woman to the operating theatre. Call the Consultant Obstetrician and consultant Anaesthetist if not on the labour ward already.

3.6.3 In Theatre

- The on call specialist Obstetrician and Anaesthetist must be made aware of any patient being transferred to the theatre from the labour ward. Continue bimanual compression of the uterus.
- Perform an examination under anaesthesia. Correct any correctible contributors (e.g. remove retained products or repair lacerations)

3.6.4 If bleeding due to atony persists

- Continue bimanual compression of the uterus and continue fluid replacement.
- Carboprost 250 micrograms IM may be given at 15 minutely intervals up to a total of 8 doses (2mg) (including the dose if given on the labour ward)
- Consider a repeat dose of Tranexamic Acid 30 minutes after the initial dose. If not given already, give the primary bolus dose in theatre. (one gram slow IV injection over 10 minutes).

3.6.5 Consider

- Inserting an arterial line
- Administration of antibiotics
- Continue bimanual compression of the uterus and consider moving to another modality of treatment.

3.6.6 Retained Placenta

- If the Placenta is retained, remove it as soon as possible under appropriate anaesthesia. See clinical guideline [Birth - Third Stage of Labour – Active Management](#) for Care of women with a Retained Placenta.

3.6.7 Genital Trauma

- Check the genital tract for presence of trauma/lacerations and repair appropriately.

3.6.8 Uterine/Vaginal Tamponade: consider balloon tamponade before surgical options:

- Insert an intrauterine balloon to effect tamponade of the placental bed. The balloon is placed either digitally or under ultrasound. The balloon is inflated with saline, using a 50ml syringe until resistance is encountered, or from a 500ml bag using gravity and a giving set. Placement of the balloon may be checked with ultrasound. A variety of balloons are available and at AHS, Bakri is available in the operating theatre.

3.6.9 Aortic Tamponade

- Consider applying aortic compression at surgery or via pressure through the abdominal wall (This may be helpful as a temporary measure if the patient is in shock).

3.6.10 Laparotomy/Surgical Therapy-consider calling a second consultant for help

For management of PPH that is unresponsive to medical therapy

- B-Lynch suture or modified B-Lynch suture
- Pelvic vessel ligation
- Hysterectomy

3.6.11 Transfer to Intensive Care Unit or Tertiary Care Hospital

- Consider admission to ICU if:
 - PPH is severe.
 - If deemed necessary by Specialist Obstetrician and/or Anaesthetist after consultation
 - if a Bakri Balloon has been inserted.
- If the woman has a blood loss greater than 3000ml and/or requires ventilation, transfer out of AHS will be considered. The Specialist Obstetrician, Anaesthetist and/or ICU Consultant will communicate with Tertiary hospital to make arrangements.

Offer explanation before discharging patients after major PPH in the ward at next available opportunity and 6-8 weeks postpartum with the consultant involved in the management of PPH.

4. Legislative context

The following documents are required to give effect to this policy [i.e. the documents included are mandatory]:

- Health Department of Western Australia (HDWA) [Clinical Handover Policy MP 0095](#)
- HDWA [Consent to Treatment Policy](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0086/18](#)
- East Metropolitan Health Service (EMHS) [Consent Policy EMHS: 5](#)

5. Supporting information

The following documents support the implementation of this policy:

- [Birth – Third Stage of Labour – Active Management Procedure](#)
- [Obstetric and Neonatal Emergency Response Procedure](#)
- [Recognition and Response Acute Deterioration](#)

6. Compliance, monitoring and evaluation

Compliance against this guideline will be evaluated through monitoring of reported clinical incident data via Datix CIMS.

Specific findings are to be reported at the relevant departmental meeting for review and identification of any opportunities for improvement.

7. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 5 – Action 5.10: Clinicians use relevant screening processes:
 - a. On presentation, during clinical examination and history taking, and when required during care
 - b. To identify cognitive, behavioural, mental and physical conditions, issues, and risks of harm
 - c. To identify social and other circumstances that may compound these risks
- Standard 6 – Action 6.8: Clinicians use structured clinical handover processes that include:
 - a. Preparing and scheduling clinical handover
 - b. Having the relevant information at clinical handover
 - c. Organising relevant clinicians and others to participate in clinical handover
 - d. Being aware of the patient's goals and preferences
 - e. Supporting patients, carers and families to be involved in clinical handover, in accordance with the wishes of the patient
 - f. Ensuring that clinical handover results in the transfer of responsibility and accountability for care
- Standard 7: Action 7.5: Clinicians document decisions relating to blood management, transfusion history and transfusion details in the healthcare record
- Standard 7: Action 7.6: The health service organisation supports clinicians to prescribe and administer blood and blood products appropriately, in accordance with national guidelines and national criteria

- Standard 8: Action 8.4: The health service organisation has processes for clinicians to detect acute physiological deterioration that require clinicians to:
 - a. Document individualised vital sign monitoring plans
 - b. Monitor patients as required by their individualised monitoring plan
 - c. Graphically document and track changes in agreed observations to detect acute deterioration over time, as appropriate for the patient
- Standard 8: Action 8.6: The health service organisation has protocols that specify criteria for escalating care, including:
 - a. Agreed vital sign parameters and other indicators of physiological deterioration
 - b. Agreed indicators of deterioration in mental state
 - c. Agreed parameters and other indicators for calling emergency assistance
 - d. Patient pain or distress that is not able to be managed using available treatment
 - e. Worry or concern in members of the workforce, patients, carers and families about acute deterioration

8. References

1. [NG235 Intrapartum care: Evidence review O 03/12/2024 \(nice.org.uk\)](https://www.nice.org.uk/NG235)
2. World Health Organization. WHO recommendations for the prevention and treatment of postpartum haemorrhage. Geneva, Switzerland: WHA Press;2012.
3. RCOG Postpartum Haemorrhage, Prevention and Management (Green-top 52)2016
4. RANZCOG College Statement: C-Obs 43 Management of Postpartum Haemorrhage (PPH)
5. King Edward Hospital, Western Australia, Management and prevention of Primary Postpartum Haemorrhage.
6. Lancet 2017;389:2105-16 Effect of early tranexamic acid administration on mortality, hysterectomy, and other morbidities in women with post-partum haemorrhage (WOMAN):an international, randomised, double-blind, placebo-controlled trial.
7. KEMH Clinical Guideline, O&M, Intrapartum: Third Stage: Active Management Following a Vaginal Birth

9. Glossary

The following definitions are relevant to this policy.

Term	Definition
Primary PPH	Occurs within the first 24 hours of birth
Secondary PPH	Occurs after 24 hours and within 12 weeks post-partum
PPH	World Health Organisation defines PPH as vaginal bleeding in excess 500 ml post childbirth irrespective of the mode of delivery.¹ This definition depends upon subjective observation. To reduce the misdiagnosis of PPH objective assessment parameters are used. Any blood loss postpartum should be considered significant if: • The woman is hemodynamically unstable • The woman has had > 10% fall in haematocrit between admission and the postpartum period

Term	Definition
	Requires a transfusion of red blood cells

10. Document owner

Head of Department Obstetrics and Gynaecology

Enquiries relating to this guideline may be directed to:

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11. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

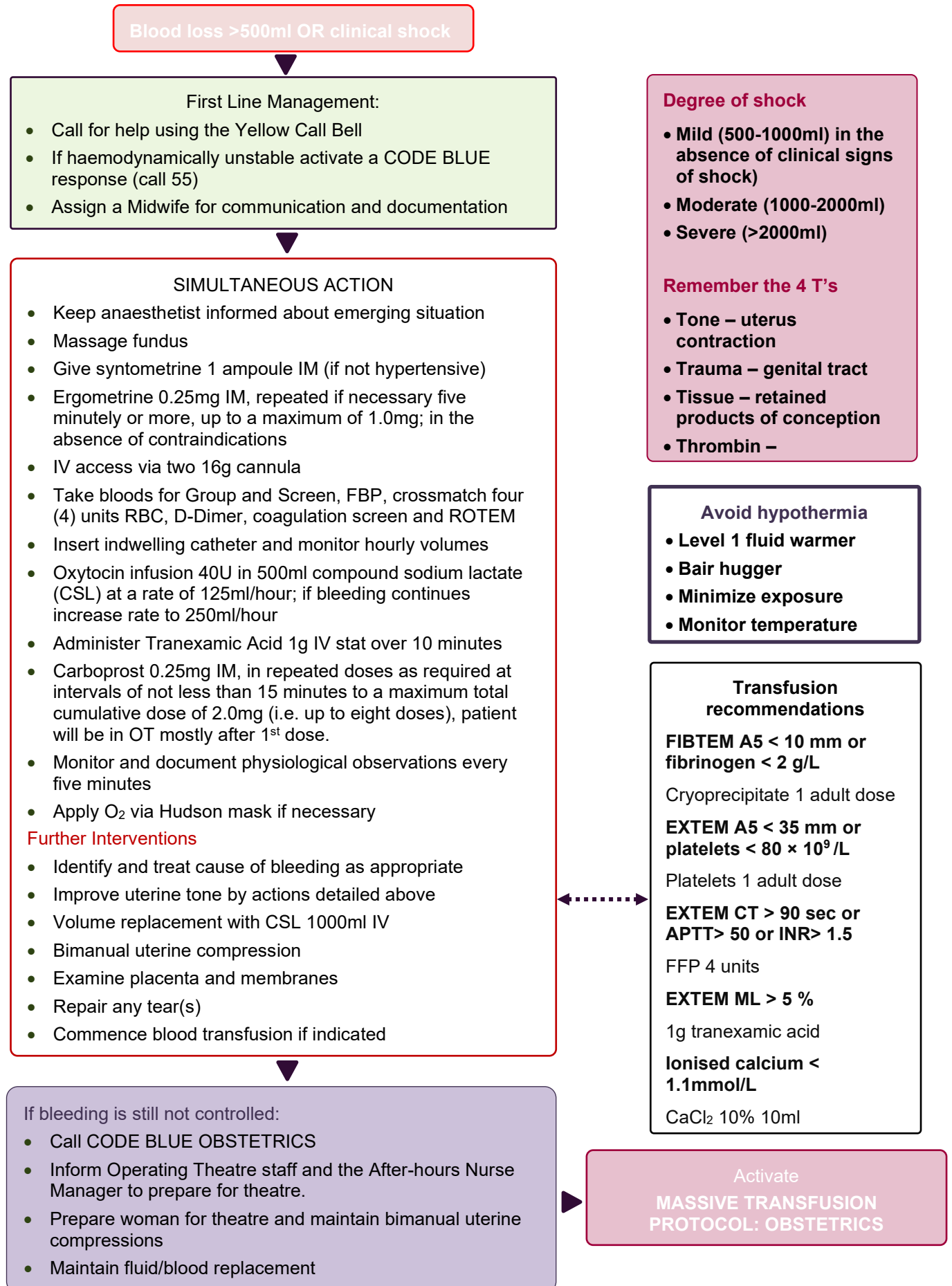
Version	Effective from	Effective to	Amendment(s)
V4			Scheduled review

12. Authorisation

This guideline has been authorised and issued by the Director Clinical Services

Authorised by	Dr Tania Strickland – Director Clinical Services
Authorisation date	28/02/2025
Published date	05/03/2025

Appendix 1: Post-partum haemorrhage flowchart



Appendix 2: Massive Transfusion Protocol Obstetrics

Red cell availability (Approximate times):

Immediate – Emergency uncrossmatched O Rh (D) negative

40 min – Crossmatched

FFP availability

30min

Cryoprecipitate availability

10min

Platelet availability

>1 hour

Fibrinogen concentrate availability

Immediately (ICU/pharmacy)

Declare a major hemorrhagic event if any of the following is present:

- Hemorrhagic shock
- Estimated blood loss >1500ml
- Laboratory or clinical evidence of coagulopathy
- Four units of Red Blood Cells transfused and more anticipated

Baseline:

Full blood count, coagulation screen, ROTEM, Group & Screen, ABG

Notify laboratory (extension: 2028/2030) to 'Activate MTP':

Delegate one person to communicate with the laboratory

Obstetric hemorrhage consideration:

Early administration of Tranexamic Acid (1g) IV and cryoprecipitate (2 adult doses)

No ROTEM

MT Pack 1:

- 4 units RBC
- 2 units FFP
- 1 adult dose fibrinogen

MT Pack 2:

- 4 units RBC
- 2 units FFP
- 1 adult therapeutic dose platelets

Continued bleeding?

FSH Haematologist to be consulted for advice

ROTEM

Senior Clinician requests:

- 4 units RBC
- Blood components guided by ROTEM
- Repeat ROTEM 10 mins after blood components given

Bleeding controlled?

Notify transfusion laboratory to: 'Cease MTP'

MONITOR

(every 30-60 mins):

- Full blood count
- Coagulation screen
- Ionized calcium
- Arterial blood gases

AIM FOR:

- Temperature >36°C
- pH > 7.35
- Lactate < 4mmol/L
- Ca²⁺ > 1.1mmol/L
- Platelets > 50x10⁹/L

Transfusion recommendations

FIBTEM A5 < 10 mm or fibrinogen < 2 g/L

Cryoprecipitate 1 adult dose

EXTEM A5 < 35 mm or platelets < 80 × 10⁹/L

Platelets 1 adult dose

EXTEM CT > 90 sec or APTT > 50 or INR > 1.5

FFP 4 units

EXTEM ML > 5 %

1g tranexamic acid

Ionised calcium <

Dosage

1 adult dose fibrinogen

Cryoprecipitate (8 units) = 2g fibrinogen concentrate

Obstetric interventions:

- Identify and treat cause of bleeding as appropriate
- Uterotonic agents (e.g. oxytocin, ergometrine)
- Physical measures (e.g. uterine compression)
- Examination under anaesthesia and repair
- Intrauterine tamponade (e.g. Bakri balloon)
- B-Lynch suture
- Consider hysterectomy

General advice:

- Keep patient warm
- Avoid excess colloid solutions
- Consider cell salvage